Zeyu Yang

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Research Interests

AI Efficiency (Quantization, Parameter-Efficient Fine-Tuning, Lossless Compression); AI Agents (Self-Evolving, Long-Horizon Planning, Multi-Agent Collaboration); AI Safety (Fairness, Privacy)

Education

Rice University August 2024 – Present
Ph.D. in Electrical and Computer Engineering, Advisor: Prof. Anshumali Shrivastava

Rice University August 2021 – May 2023
M.S. in Data Science, Advisor: Prof. Joe Warren and Prof. Akane Sano

University of Electronic Science and Technology of China August 2017 – May 2021
B.E. in Electronic Information Engineering, GPA: 3.66/4.00

Industry Experience

Meta Superintelligence Lab August 2025 – Present
Visiting Researcher, with Anshumali Shrivastava and Jason Chen

- [arXiv'26] Developed SIRA, a retrieval agent that compresses multi-round exploratory search into a single corpus-discriminative BM25 call by using LLM-predicted expansion terms validated against document-frequency statistics, significantly outperforming dense retrievers and multi-round agentic baselines across ten BEIR benchmarks while remaining training-free and interpretable [8].

Research Experience

Rice Rush Lab December 2024 – Present
Graduate Research Assistant, with Prof. Anshumali Shrivastava

- **[ICLR'26]** Developed an entropy-based lossless compression algorithm for float8 neural network parameters, achieving an 18% compression ratio while maintaining bit-exact reconstruction of the original model [3].
- **[ICML'26]** Contributed to Sketch&Walk Attention, a training-free sparse attention method that uses Hadamard sketching to approximate attention scores and a cross-layer walk mechanism to select top-k attention blocks, achieving up to 6× inference speedup while maintaining near-lossless accuracy at 20% attention density [2].
- Proposed a compressed adaptation framework for diffusion transformers that performs global clustering of weights and represents each weight via a global cluster ID and an index into that cluster's lookup table, matching or exceeding full-precision LoRA while substantially outperforming QLoRA in quality, training time, and memory across 6 diffusion transformers (2B–20B) [1].

Rice Computational Wellbeing Group January 2024 – July 2024
Graduate Student Researcher, with Prof. Akane Sano

- **[TMLR'25]** Developed a tabular diffusion model with sensitive guidance that generates fair synthetic data with balanced joint distributions of target labels and sensitive attributes, outperforming existing methods on fairness metrics such as demographic parity ratio and equalized odds ratio by over 10% [7].
- [CHIL'25] Designed a self-supervised contrastive pretraining method that jointly learns from multimodal wearable sensor time series and survey data, improving stress detection accuracy while reducing reliance on labeled data [4].

Rice Computer Graphics/Geometric Design Group May 2022 – August 2022
Graduate Student Researcher, with Prof. Joe Warren

- [GitHub'22] Developed a Python toolbox for animating the training progress of multi-layer perceptrons [10].

Monash Data Futures Institute December 2020 – March 2021

Undergraduate Student Researcher, with Prof. Hao Wang

- [ARXIV'23] Developed an asynchronous distributed alternating direction method of multipliers (ADMM) algorithm to optimize energy trading problems under asynchronous communication, allowing communication delay and indicating a potential for better outcomes in real-world applications [9].

Selected Publications [Full List]

In Submission:

[1] **Zeyu Yang**, Tianyi Zhang, Junda Su, Yang Sui, Anshumali Shrivastava, "Globally-Sensitive Sketch for Compressed Adaptation of Diffusion Transformers", 2026.

Conference Publications:

[2] Hoang Anh Duy Le, Sahil Joshi, **Zeyu Yang**, Zhaozhuo Xu, Anshumali Shrivastava, "Scout Before You Attend: Sketch-and-Walk Sparse Attention for Efficient LLM Inference", in *International Conference on Machine Learning (ICML)*, 2026.

[3] **Zeyu Yang**, Tianyi Zhang, Jianwen Xie, Chuan Li, Zhaozhuo Xu, Anshumali Shrivastava, "To Compress or Not? Pushing the Frontier of Lossless GenAI Model Weights Compression with Exponent Concentration", in *International Conference on Learning Representations (ICLR)*, 2026.

[4] **Zeyu Yang**, Han Yu, Akane Sano, "Contrastive Pretraining for Stress Detection with Multimodal Wearable Sensor Data and Surveys", in *Conference on Health, Inference, and Learning (CHIL)*, 2025.

[5] Yuanhao Gong, Tan Tang, **Zeyu Yang**, Lantao Yu, "A Filter for Minimizing Gaussian Curvature on 3D Triangular Meshes", in *International Symposium on Biomedical Imaging (ISBI)*, 2025.

[6] Yizhuo Yang, Huan Wang, Zhiliang Liu, **Zeyu Yang**, "Few-Shot Learning for Rolling Bearing Fault Diagnosis via Siamese Two-Dimensional Convolutional Neural Network", in *Asia-Pacific International Symposium on Advanced Reliability and Maintenance Modeling*, 2020.

Journal Publications:

[7] **Zeyu Yang**, Han Yu, Peikun Guo, Khadija Zanna, Xiaoxue Yang, Akane Sano, "Balanced Mixed-Type Tabular Data Synthesis with Diffusion Models", *Transactions on Machine Learning Research (TMLR)*, 2025.

Preprints:

[8] **Zeyu Yang**, Qi Ma, Jason Chen, Anshumali Shrivastava, "Superintelligent Retrieval Agent: The Next Frontier of Information Retrieval", *arXiv:2605.06647*, 2026.

[9] **Zeyu Yang**, Hao Wang, "Network-Aware Asynchronous Distributed ADMM Algorithm for Peer-to-Peer Energy Trading", *arXiv:2312.06976*, 2023.

Published Software:

[10] *PlotNet*. (2022). [Online]. Available: https://github.com/zeyuyang8/plotnet

Awards & Honors

Outstanding Undergraduate Student Award	2021
Meritorious Winner of Interdisciplinary Contest in Modeling	2020

Skills

Programming:

- *Data Science:* C++, Rust, CUDA, Python (PyTorch, SGLang, Hugging Face, Weights & Biases, Matplotlib), MATLAB, Tableau
- *Web Development:* Python (Django, Dash), HTML, CSS, JavaScript (React), PostgreSQL
- *IT Skills:* Git, AWS, Docker, Linux, Large-Scale Multi-Node Training and Inference, Markdown, LaTeX, Shell Scripting

Machine Learning:

- *Deep Learning:* Large-Language Models, Diffusion Models, Transformers, Contrastive Learning
- *Machine Learning:* Information Retrieval, BM25, Sparse Dictionary Learning, Entropy Coding, Matrix Factorization, Feature Selection, Bias Mitigation

Coursework

Graduate Coursework:

- *Computer Science:* Programming for Data Science, Big Data Management for Data Science, Data Visualization, Graduate Design and Analysis of Algorithms
- *Machine Learning:* Statistical Machine Learning, Deep Learning for Vision and Language, Applied Machine Learning and Data Science Projects, AI for Health, Advanced Machine Learning, Learning from Sensor Data
- *Statistics & Optimization:* Statistics for Data Science, Convex Optimization, Information Theory

Undergraduate Coursework:

- *Computer Science:* Introductory C Programming, Introductory Python Programming
- *Electrical Engineering:* Application and Design of Digital Logic, Signals and Systems, Digital Communication, Microelectronic Systems, Circuit Analysis and Design, Fundamentals of Analog Circuits
- *Math & Physics:* Linear Algebra and Space Analytic Geometry, Probability Theory and Mathematical Statistics, Calculus I, Calculus II, Physics I, Physics II

Teaching Experience

Introduction to Data Science (DSCI 101) at Rice University, Fall 2025
Teaching Assistant, with Prof. Lorenzo Luzi

Data Visualization (COMP 665) at Rice University, Spring 2022 – Spring 2023
Teaching Assistant, with Prof. Joe Warren

Statistics for Data Science (COMP 680) at Rice University, Spring 2023
Academic Tutor, with Prof. Chen Su